

	through the slits. If the wavefronts do not overlap at the screen, there will be no interference, hence fringes are not observed.
3(a)	p.d. across $1.5\text{ k}\Omega$ resistor = $(1.60 \times 10^{-3})(1.50 \times 10^3) = 2.40\text{ V}$ p.d. across thermistor = $6.00 - 2.40 = 3.60\text{ V}$ Resistance of thermistor = $\frac{3.60}{(1.60 \times 10^{-3})} = 2.25 \times 10^3\ \Omega$ (using $V = IR$) From Fig. 3.2, temperature is $14\text{ }^\circ\text{C}$.
3(b)	As current flows through the thermistor, there will be a heating effect. This increases the temperature of the thermistor. From Fig 3.2, as temperature increases, the resistance of the thermistor decreases, decreasing the total resistance of the circuit and increasing current flow through the thermistor (since emf remains the same). This will further increase the temperature of the thermistor if it is not kept cooled.
4(a)	One tesla is the magnetic flux density when a conductor of length 1 m